ROOTEDAI

Software Development Plan

Adhi Compliance — AI Compliance, Automated.

FULL DEVELOPMENT PLAN

RootedAI — Engineering Intelligence & Enterprise Automation

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1. Project Overview & Objectives

Adhi Compliance is an AI-powered compliance management platform that enables organizations deploying AI systems to achieve and maintain regulatory compliance across multiple jurisdictions — EU AI Act, US state laws, India's DPDP Act, and China's AIGC provisions.

Primary Objectives

1. **Build a production-ready SaaS platform** that auto-classifies AI systems by risk level and monitors compliance in real-time
2. **Launch an MVP within 6 months** targeting Indian IT companies serving EU/US clients as the initial market
3. **Establish RootedAI** as a first-mover in the India-focused AI compliance space
4. **Generate recurring revenue** through tiered SaaS subscriptions (₹50K–₹2L+/month)
5. **Create defensible IP** around regulation-as-data pipelines and compliance AI agents

Product Vision

Adhi Compliance is the **compliance layer every AI company plugs into**. One platform for risk classification, continuous monitoring, bias auditing, and regulatory reporting. Think of it as "Datadog for AI Compliance" — a horizontal platform that becomes essential infrastructure for any organization deploying AI.

> **Pivot Context:** Adhi Compliance is being pivoted from a multimodal RAG system. The existing tech stack (FastAPI, Next.js, Celery/Redis, FAISS, Gemini) and features (multi-org support, file ingestion, RAG query engine) form the foundation. This plan covers the delta from current state to full compliance platform.

2. Scope Definition

In Scope (v1.0)

- AI System Registry — register, catalog, and manage AI systems per organization
- Risk Classification Engine — rule-based + LLM-powered EU AI Act risk classification
- Regulation Knowledge Base — structured, searchable database of AI regulations (EU, US, India, China)
- Compliance Chat (RAG) — natural language Q&A against regulation corpus
- Compliance Dashboard — executive overview with scores, alerts, and status tracking

- Bias Auditor — upload datasets, run fairness metrics, generate audit reports
- Model Card Generator — auto-generate model cards from system metadata
- Incident Reporting — EU AI Act Article 62 workflow with deadline tracking
- Continuous Monitoring — scheduled compliance re-checks with alerting
- Report Generation — PDF/Excel compliance reports and model cards
- Multi-tenant SaaS — company isolation, user management, subscription tiers
- Webhook Integrations — Slack, Teams, Email notifications

Out of Scope (v1.0)

- Mobile native apps (PWA may be considered post-launch)
- Real-time model monitoring (connecting to live ML inference pipelines)
- Automated regulatory filing/submission to authorities
- White-label / OEM distribution
- Non-English regulatory document processing (v2.0 — Sarvam AI for Indian languages)
- SOC 2 / ISO 27001 certification (post-launch priority)

3. Team Structure & Roles

Role	Person	Responsibilities	Allocation
Founder / Lead Dev	Ganesh Khovalan	Architecture, backend, AI/ML, product decisions, project management	Full-time
AI Coding Assistant	Claude / Cursor / Copilot	Code generation, reviews, testing, documentation	Always-on
Frontend (Contract)	TBD (Phase 2)	Next.js UI implementation from designs	Part-time
Design (Contract)	TBD (Phase 1)	UI/UX wireframes, design system, Figma	2–3 weeks
Legal Advisor	TBD	Regulatory accuracy review, compliance content validation	Advisory

> **Solo Dev + AI Model:** This plan is designed for a solo developer augmented by AI coding assistants. Sprint velocities are calibrated for this model — roughly 2x what a solo dev could achieve unassisted. Critical path items are serialized; parallelizable work leverages AI pair-programming.

4. Development Methodology

- **Framework:** Agile Scrum (adapted for solo dev)
- **Sprint Length:** 2 weeks
- **Ceremonies:** Weekly self-review (Monday), sprint planning (start), sprint retro (end)
- **Tracking:** GitHub Projects (Kanban board) with Issues linked to PRs
- **Branching:** GitFlow — `main` (prod), `develop` (staging), `feature/*` branches
- **CI/CD:** GitHub Actions — lint, test, build, deploy on merge
- **Documentation:** Inline (JSDoc/docstrings) + Architecture Decision Records (ADRs) for major choices
- **Quality Gates:** Every PR requires: passing tests, type-check clean, no lint errors

5. Phase 1: Planning & Foundation (Weeks 1–3)

PHASE 1 Duration: 3 weeks · Focus: Architecture, Design, Setup

Week 1: Requirements & Architecture

- Finalize product requirements document (PRD) with feature prioritization (MoSCoW)
- Complete architecture design (reference Architecture Document v1.0)
- Define complete database schema — all tables, relationships, indexes
- Design API contracts — OpenAPI 3.0 specification for all endpoints
- Select and configure Supabase project (PostgreSQL + Auth + Storage)
- Set up pgvector extension and vector index

Week 2: UI/UX Design & Dev Environment

- Create wireframes for all primary pages (Figma or Excalidraw)
- Define design system tokens: colors, typography, spacing, component library
- Set up monorepo structure (or separate repos) for frontend + backend
- Initialize Next.js 16 project with TypeScript, Tailwind, Framer Motion
- Initialize FastAPI project with SQLAlchemy, Alembic, Celery
- Configure Docker Compose for local development
- Set up GitHub repos, branch protection, CI/CD pipelines

Week 3: Foundation Code

- Implement Supabase Auth integration (frontend + backend JWT validation)
- Create SQLAlchemy models for all core tables
- Run initial Alembic migrations
- Implement RLS policies for multi-tenant isolation
- Build API middleware stack: auth, rate limiting, CORS, error handling
- Create shared UI component library: Button, Card, Input, Modal, Table, Badge
- Build dashboard layout shell: sidebar, topbar, main content area

Phase 1 Deliverables

Deliverable	Status Criteria
Architecture Document	Reviewed and finalized
OpenAPI Specification	All endpoints defined with schemas
Database Schema	All tables created, RLS active
Dev Environment	Docker Compose up, all services running
Auth System	Register/login/JWT working end-to-end
UI Shell	Dashboard layout with navigation working

6. Phase 2: Core Platform Build (Weeks 4–10)

PHASE 2 Duration: 7 weeks (4 sprints) · Focus: Core Features

7. Phase 3: Advanced Features (Weeks 11–16)

PHASE 3 Duration: 6 weeks (3 sprints) · Focus: Differentiation Features

8. Phase 4: Enterprise & SaaS (Weeks 17–22)

PHASE 4 Duration: 6 weeks (3 sprints) · Focus: Productization

9. Phase 5: Testing & QA (Weeks 23–25)

PHASE 5 Duration: 3 weeks · Focus: Quality Assurance

Unit Testing

- Backend: pytest with 80%+ coverage target on services and compliance engine
- Frontend: Vitest + React Testing Library for component tests
- API contract tests against OpenAPI spec

Integration Testing

- End-to-end flows: registration → system creation → classification → compliance check → report
- Celery task integration tests (regulation loading, bias audits, report generation)
- Webhook delivery testing with mock endpoints
- Multi-tenant isolation verification

Security Audit

- OWASP Top 10 review: injection, XSS, CSRF, broken auth, misconfig
- RLS policy audit: verify no cross-tenant data leakage
- JWT handling review: token storage, refresh flow, expiry
- Dependency vulnerability scan (Snyk/Dependabot)
- API rate limiting and abuse testing

Performance Testing

- Load testing: 100+ concurrent users, 1000+ API requests/minute
- Vector search performance: query latency <200ms at 100K chunks
- LLM response times: classification <10s, chat <5s first token
- Report generation: PDF <30s, Excel <15s
- Database query optimization (EXPLAIN ANALYZE on critical queries)

User Acceptance Testing

- Recruit 5–10 beta testers from Indian IT companies
- Structured feedback forms per feature area

- Bug triage and priority fix cycle
- UX iteration based on user feedback

10. Phase 6: Deployment & Launch (Weeks 26–28)

PHASE 6 Duration: 3 weeks · Focus: Go Live

Week 26: Staging Environment

- Deploy full stack to AWS staging environment
- Configure Supabase production project (separate from dev)
- Set up Redis (ElastiCache) for production
- Environment variable management via AWS Secrets Manager
- SSL certificate configuration
- Smoke testing on staging

Week 27: Production Deployment

- DNS setup: compliance.rootedai.co.in (or adhi.rootedai.co.in)
- Deploy backend to ECS Fargate (2+ FastAPI instances, 2+ Celery workers, 1 Beat)
- Deploy frontend to Vercel with custom domain
- Configure CloudFront CDN
- Set up monitoring: Sentry (errors), LogRocket (sessions), CloudWatch (infra)
- Alerting: PagerDuty/Slack for production errors
- Database backups: daily automated snapshots

Week 28: Launch

- Final production smoke test
- Seed production regulation database (EU AI Act, NIST, DPDP, AIGC)
- Create marketing landing page
- Prepare demo video and documentation
- Launch checklist sign-off
- 🚀 **Public beta launch**

Launch Checklist

Item	Status
All critical bugs resolved	☐
Security audit passed	☐
Performance benchmarks met	☐
Monitoring & alerting active	☐
Backup & recovery tested	☐
SSL/TLS configured	☐
Terms of Service & Privacy Policy published	☐
Documentation complete	☐
Billing system tested	☐
4+ regulations loaded and validated	☐

11. Phase 7: Post-Launch (Weeks 29+)

PHASE 7 Ongoing · Focus: Growth & Iteration

Beta User Onboarding (Weeks 29–32)

- Onboard 10–20 beta companies (free/discounted tier)
- Weekly feedback calls with early adopters
- Bug fix sprints (1-week cycles)
- Feature prioritization based on user feedback

Iteration Cycles (Weeks 33+)

- Bi-weekly feature releases based on feedback
- Regulation database expansion (US state laws, sector-specific regulations)
- Performance optimization based on production metrics
- Multi-language support (Hindi, Tamil via Sarvam AI integration)

Marketing Support

- Content marketing: blog posts on AI compliance topics
- Webinars: "EU AI Act Compliance for Indian IT Companies"
- Conference presence: NASSCOM, AI/ML meetups
- LinkedIn thought leadership from founder
- Case studies from beta customers

12. Risk Register

Risk	Impact	Probability	Mitigation
Solo dev burnout	🔴 High	Medium	Strict sprint boundaries, AI pair-programming, consider contractor for frontend
Regulatory accuracy	🔴 High	Medium	Legal advisor review, disclaimers ("not legal advice"), regular content audits
LLM hallucination	🔴 High	Medium	RAG with source citations, confidence scoring, human review for critical outputs
Gemini API changes/pricing	🟡 Medium	Low	Abstract LLM layer, easy swap to Claude/GPT-4, cache frequent queries
Supabase limits	🟡 Medium	Low	Pro plan from launch, migration path to self-hosted PostgreSQL if needed
Slow market adoption	🟡 Medium	Medium	Freemium tier, content marketing, conference presence, build in public
Regulation changes	🟢 Low	High	This IS the product — continuous regulation updates are a feature
Security breach	🔴 High	Low	RLS, encryption, security audit, minimal data collection, bug bounty post-launch
Competitor launch	🟡 Medium	Medium	Speed to market, India-first pricing, unique multimodal heritage
Scope creep	🟡 Medium	High	Strict scope definition, MVP mindset, defer non-core features to v2

13. Budget Estimate

Monthly Infrastructure Costs (Post-Launch)

Service	Tier	Monthly Cost

Supabase	Pro	$25
AWS ECS Fargate (2 services)	0.5 vCPU, 1GB each	~$70
AWS ElastiCache Redis	cache.t3.micro	~$15
Vercel	Pro	$20
CloudFront CDN	Usage-based	~$10
Google Gemini API	Pay-as-you-go	~$50–200
Sentry	Team	$26
Domain + SSL	Annual	~$5
SendGrid (email)	Free/Essentials	$0–20
Total Monthly		**$220–390**

One-Time / Development Costs

Item	Estimate
UI/UX Design (contract)	₹50K–1L
Frontend developer (contract, 2 months)	₹1–2L
Legal review of compliance content	₹50K–1L
AI coding tools (Cursor Pro, etc.)	$20/mo
Total One-Time	**₹2–4L ($2.5–5K)**

14. Milestones & Timeline

Week	Milestone	Phase	Key Deliverable
Week 3	**Foundation Complete**	P1	Auth working, DB schema live, UI shell ready
Week 5	**AI System Registry Live**	P2	Full CRUD for companies + AI systems
Week 7	**Regulation Engine Live**	P2	4 regulations loaded, RAG chat working
Week 9	**Risk Classifier Live**	P2	Auto-classification + compliance checks
Week 10	**MVP Dashboard**	P2	Full dashboard with live data — internal demo
Week 12	**Bias Auditor Live**	P3	Bias audits + model card generation
Week 14	**Monitoring Active**	P3	Continuous monitoring + webhook alerts
Week 16	**Feature Complete**	P3	All v1.0 features built — incidents, reports, cross-border
Week 18	**SaaS Ready**	P4	Billing, tiers, onboarding flow
Week 22	**Enterprise Ready**	P4	API keys, CI/CD integration, agent monitor
Week 25	**QA Complete**	P5	All tests passing, security audit cleared
Week 28	🚀 **PUBLIC BETA LAUNCH**	P6	Production deployment, first users
Week 32	**10+ Beta Companies**	P7	Validated product-market fit

15. Success Metrics

Phase	KPI	Target
Phase 1	Development velocity	All foundation tasks completed in 3 weeks
Phase 2	Feature completeness	4 core features working end-to-end
	Classification accuracy	≥85% agreement with manual expert classification
	RAG answer quality	≥80% factually correct with proper citations
Phase 3	Feature completeness	All v1.0 features built and integrated
	Bias audit coverage	4+ fairness metrics implemented
Phase 4	SaaS readiness	Billing working, 3 tiers configured
Phase 5	Test coverage	≥80% backend, ≥60% frontend
	Security	Zero critical/high vulnerabilities
	Performance	P95 API latency <500ms, vector search <200ms
Phase 6	Uptime	99.5% in first month
Phase 7	Beta users	10+ companies in first 4 weeks
	NPS	≥40 from beta users
	Conversion	≥20% free → paid within 60 days

16. Dependencies & Assumptions

Dependencies

- **Supabase** — platform availability and pgvector support stability
- **Google Gemini API** — continued availability and reasonable pricing for 1.5 Flash
- **Regulatory content** — access to full text of EU AI Act, NIST AI RMF, India DPDP, China AIGC
- **AWS** — ECS Fargate, ElastiCache, Secrets Manager availability in ap-south-1
- **Design contractor** — availability during Phase 1 (Weeks 2–3)

Assumptions

- Solo developer (founder) available full-time for the 28-week build period
- AI coding assistants provide 1.5–2x productivity multiplier consistently
- EU AI Act requirements remain stable through development (major provisions effective Aug 2025)
- Indian IT companies serving EU/US clients represent a viable initial market
- ₹50K–₹2L/month pricing is acceptable for the target market
- Legal disclaimer ("not legal advice") is sufficient for v1.0; formal legal review before enterprise sales
- Open-source bias/fairness libraries (AIF360, Fairlearn) provide sufficient algorithmic foundation
- Infrastructure costs remain within $220–400/month range at launch scale